

Factors Affecting Dissolving

- Dissolving (at the molecular level) occurs when the solute is freed from the intermolecular attractions holding it in its solute lattice.
- If the bond and attractions within the solute are too strong, the solvent intermolecular attractions will not be able to remove the solute from the lattice and the solute does not dissolve well.
- If the intermolecular attractions between solvent and the solute are stronger than the attractions/bonds within the solute, then the solvent will rip part of the solute away and surround it; hence, part of the solute has dissolved.
- We can physically change the collective amount of intermolecular attractions between the solute and the solvent and this is typically how we improve dissolving:

1. Stirring

Stirring increases the amount of solute/solvent contact which means more solvent is forming attractions to solute. So collectively, the solvent is more effective at dissolving the solute allowing more and more solute to dissolve.

2. Changing the Surface Area

Grinding the solute into smaller pieces is also an intuitive method of improving dissolving. This increases the surface area of the solute exposing more of it to the solvent. This again increases solute/solvent interactions meaning more solute will be drawn away into solution by the solvent.

3. Changing the Temperature

Increasing the temperature is also an intuitive method to improve the rate of dissolving. Generally, a higher temperature imparts more kinetic energy to both solute and solvent which increases their molecular motion so there will be more solute/solvent interactions resulting in more solute dissolving.

For most substances: $A_{(s,l,g)} + \text{heat} \rightarrow A_{(aq)}$

In this case, the addition of heat will result in more products and hence more solute dissolves. This is because the heat is not only allowing more solute/solvent interaction through increased molecular motion, but is also breaking the lattice interactions within the solute causing it to disintegrate.

However, for some solids and liquids and for all gas solutes, the process of dissolving is actually exothermic according

to: $A_{(g)} \rightarrow A_{(aq)} + \text{heat}$

In this case, the addition of heat actually inhibits the reaction so dissolving is reduced as temperature increases. For gases, hotter temperatures causes them to spread out more (as well as move faster) as the intermolecular attractions (not strong to begin with) between particles weakens with heat and so the gas escapes the solvent faster decreasing solute/solvent interaction time and less total solute dissolved.

For example:

- The dissolved carbon dioxide gas comes out of solution as it warms up.
- Many types of fish prefer colder water (higher dissolved oxygen content).

An important note: There are two energy changes that occur in the process of creating a solution

1. The external temperature change that affects dissolving. This is what we are considering here
2. The internal energy change that results when the solute is dissolved (due to new forces of attraction being formed)